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My Programs



10BASIC

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Pictures for the Turtle



CHAPTER 1

Welcome

This is a computer for doing a few things carefully.

You can make pictures. You can move a turtle. You can type small programs.

When you are done with something, you can always come back.



What to Type

Getting Around

Enter	start the thing you picked
Esc	go back, on a website
Ctrl + C	stop something that will not stop
Ctrl + Alt + Backspace	go back to the menu

The Turtle

FD 50	go forward
RT 90	turn right
LT 90	turn left
REPEAT	do it again
CS	clear the screen
HOME	go to the middle
PU	pen up
PD	pen down

BASIC

RUN	make it go
LIST	show your lines
NEW	start over
CLS	clear the screen
PRINT	show something
INPUT	ask and wait
FOR	do lines again
GOTO	jump to a line
END	stop here

10.7 Name Shout

The computer asks your name!

Try this

- changing the symbols
- adding more rows

```
10 CLS
20 INPUT "WHAT IS YOUR NAME"; N$
30 FOR I = 1 TO 10
40 PRINT N$
50 NEXT I
60 END
```

Try this

- changing how many times it prints
- printing something else



CHAPTER 2

The Menu

When the computer turns on, you see the menu.

The menu lets you choose what to do.

Use the arrow keys to pick what you want. Press **Enter** to start it.

To leave and go back to the menu, hold down **Ctrl** and **Alt**, then press **Backspace**.



CHAPTER 3

Drawing

Pick “Draw Pictures” from the menu.

You get a big white page and rows of buttons around the edge.

Pick a colour. Pick a brush. Then hold the mouse button down and move the mouse across the page.

There are stamps you can press onto the page, and lines and shapes you can stretch, and an eraser for taking things away.

If you do something you did not mean to, press Undo and it goes away again.

Press Save to keep a picture. Press Open to look at the pictures you have kept.

10.5 Number Explosion

Numbers fill the screen.

```
10 CLS
20 FOR I = 1 TO 10
30 PRINT I; I; I; I; I
40 NEXT I
50 END
```

Try this

- printing more numbers
- changing `I` to `I * I`

10.6 Checkerboard

This makes a pattern.

```
10 CLS
20 FOR R = 1 TO 10
30 PRINT "* * * * *"
40 PRINT " * * * * *"
50 NEXT R
60 END
```

10.4 Bouncing Word

The word moves back and forth.

```
10 CLS
20 FOR X = 1 TO 20
30 PRINT SPC(X) + "HI!"
40 NEXT X
50 FOR X = 20 TO 1 STEP -1
60 PRINT SPC(X) + "HI!"
70 NEXT X
80 GOTO 10
```

Try changing

- "HI!" to your name
- the numbers to make it move more

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CHAPTER 4

Websites

4

Your computer can talk to other computers. It sends a message out along a wire. That wire goes to another wire, and after a great many wires the message arrives at a computer far away. All of those computers and wires together are called the Internet. Some of those far-away computers keep pages for other people to read. A page can have words on it, or pictures, or a game. When a lot of pages belong together, we call them a website. Every page has an address, the way a house has an address. Your computer uses the address to ask the far-away computer for the page, and that computer sends the page back. It does this every time you look at a page, which is why a website only works when

your computer can reach the Internet.

4.1 The Websites on This Computer

Some things on the menu are websites.

The address is already filled in for you, and there is nowhere to type a different one, so you cannot end up somewhere else by accident.

You can only visit the websites we picked together.

If the page says it is blocked

Nothing is broken. It just means that page is not one of ours. Press **Esc** to go back to the page you were reading.

If you want a new website added, ask, and we can talk about it.

10.2 Growing Stars

This makes stars grow down the screen.

```
10 CLS
20 FOR I = 1 TO 20
30 PRINT STRING$(I, "*")
40 NEXT I
50 END
```

Try changing

- **20** to a bigger number
- **"*"** to another letter

10.3 Letter Rain

Letters fall forever!

```
10 CLS
20 PRINT CHR$(65 + RND * 26);
30 GOTO 20
```

To stop it, press **Ctrl** and **C**.

If the computer does not understand a line Computers are very fussy. A missing quote mark, or a word spelled almost right, is enough to stop one. When that happens the computer prints a message and tells you which line it did not like. That is the computer trying to help, not the computer being cross. Type that line again with the same number at the front. The new line takes the place of the old one. Then type **RUN** again.

10.1 Useful Commands

RUN makes the program go

LIST shows what you typed

NEW erase and start over

CLS clear the screen

If the screen fills up or will not stop Press **Ctrl** and **C** at the same time.



CHAPTER 5

Looking Up Words

Pick "Look Up a Word" from the menu. Type a word and press **Enter**. The computer tells you what it means. If you are not sure how to spell it, type your best guess anyway. The computer will show you words that look close to what you typed, and one of them is probably the one you wanted. Press **Enter** on an empty line to go back to the menu. The words come from people all over the world who wrote them down in simple English so that anyone learning to read can understand them.



CHAPTER 6

The Clock

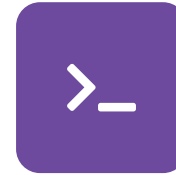
Pick “Clock” from the menu.

There are two clocks on the screen, and they always say the same thing.

One is round, with the numbers 1 to 12 around the edge and three hands in the middle. The short fat hand points at the hour. The long hand counts the minutes. The thin red one goes all the way round once every minute, so you can watch it move.

The other clock writes the time out in numbers, and the day and the date underneath.

If you are still learning the hands, read the round clock first and then look at the numbers to see if you were right.



CHAPTER 10

BASIC: Telling the Computer What to Do

You can do Daddy’s job! This computer can run small programs.

Programs are made from lines with numbers.

You type the lines. Then you type:

RUN

9.6 Useful Commands

CS	clear the screen
HOME	send the turtle back to the middle
PU	pen up (move without drawing)
PD	pen down (draw again)

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The Timer

CHAPTER 7



8

Pick “Timer” from the menu.

The computer asks how many minutes you want. Type a number and press **Enter**.

Then the big numbers count backwards, and music plays while they count.

When it gets to zero the music stops, the screen says TIME’S UP, and a bell rings until somebody stops it.

To stop the timer 

Press **Ctrl** and **C** at the same time. This works while it is counting and while the bell is ringing.



CHAPTER 8

The Stopwatch

Pick “Stopwatch” from the menu.

The timer counts backwards. The stopwatch counts forwards.

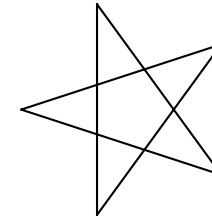
It starts at zero the moment it opens, and it keeps going until you stop it.

Use it to find out how long something takes. How long can you stand on one foot? How long does it take you to put your shoes on?

To stop it, press **Ctrl** and **C**.

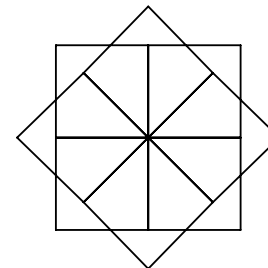
9.4 Star

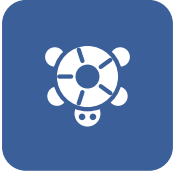
```
REPEAT 5 [FD 100 RT 144]
```



9.5 Flower

```
REPEAT 8 [REPEAT 4 [FD 50 RT 90] RT 45]
```





CHAPTER 9

The Turtle (Logo): Telling the Computer to Draw

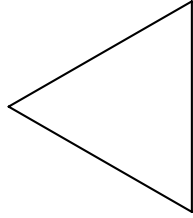
Instead of drawing yourself, you can tell the computer what to draw, so that it happens exactly like you want. The turtle listens to words. You can tell the turtle to move. You can tell it to turn. Try typing:

```
FD 50  
RT 90  
LT 90
```

10

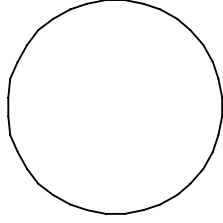
9.2 Triangle

```
REPEAT 3 [FD 100 RT 120]
```



9.3 Circle

```
REPEAT 36 [FD 10 RT 10]
```



Try this

- changing the numbers
- making it repeat more times

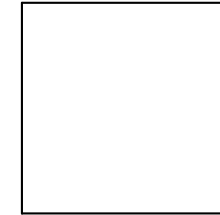
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If the turtle does not do it

The turtle knows only a few words, and it wants them spelled exactly right.

`FD 50` works. `FD50` does not, because the word and the number need a space between them.

When the turtle does not understand, it prints a message instead of drawing. Nothing is broken and you have not hurt anything. Look at what you typed, fix the one letter or the one space that is wrong, and press **Enter** again.



That is a lot of typing. You can also say:

```
REPEAT 4 [FD 100 RT 90]
```

Try changing

- `100` to make it bigger or smaller
- `90` to see what happens

9.1 Square

```
FD 100  
RT 90  
FD 100  
RT 90  
FD 100  
RT 90  
FD 100  
RT 90
```